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DETAIL SPECIFICATION

# BARIUM NITRATE

Reactivated after 28 June 2022 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the requirements, examinations, and tests for barium nitrate for use in the manufacture of propellants and other ordnance applications.

1.2 Classification. Barium nitrate is of the following Classes and Granulations.

TABLE I. Barium nitrate Classes and Granulations.

| Class | Granulation |
|---|---|
| 1 | A<br>B |
| 2 | |
| 3 | |
| 4 | |
| 5 | |
| 6 | A<br>B |

Comments, suggestions, or questions on this document should be addressed to: US Army DEVCOM Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000 or email to usarmy.pica.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil/.

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## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

| | | |
|---|---|---|
| MIL-STD-1916 | - | DOD Preferred Methods for Acceptance of Product |

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Non-Government standards and publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

AMERICAN CHEMICAL SOCIETY

| | | |
|---|---|---|
| Reagent Chemicals | - | Specifications and Procedures for Reagents and Standard Grade Reference Materials |

(Copies of this document are available from https://pubs.acs.org/.)

ASTM INTERNATIONAL

| | | |
|---|---|---|
| ASTM E11 | - | Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves |
| ASTM E300 | - | Standard Practice for Sampling Industrial Chemicals |
| ASTM B330 | - | Standard Test Methods for Estimating Average Particle Size of Metal Powders and Related Compounds Using Air Permeability |
| ASTM D1193 | - | Standard Specification for Reagent Water |

(Copies of these documents are available from https://www.astm.org/.)



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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

3.2 Color. The color of the barium nitrate shall be white. The use of any material, such as Prussian blue (ferric ferrocyanide), for the purpose of imitating a satisfactory color to the barium nitrate, shall not be permissible.

3.3 Physical and chemical properties. Barium nitrate shall conform to the physical and chemical requirements specified in Tables II and III.

TABLE II. Physical and chemical requirements for barium nitrate.

| Property | Requirements | | | | | |
|---|---|---|---|---|---|---|
| | Class 1 | Class 2 | Class 3 | Class 4 | Class 5 | Class 6 |
| Barium nitrate, weight percent minimum, (wt. % min.) | 99.0 | 99.0 | 99.5 | 99.5 | 98.5 | 99.5 |
| Strontium (as Sr), wt. % maximum, (max.) | 0.6 | - | 0.6 | - | - | 0.6 |
| Calcium (as Ca), wt. % max. | 0.05 | - | 0.05 | - | - | 0.05 |
| Iron and aluminum (as oxides), wt. % max. | - | 0.50 | - | 0.02 | - | 0.02 |
| Calcium and magnesium (as oxides), wt. % max. | - | 0.50 | - | - | - | - |
| Sodium (as Na), wt. % max. | - | 0.11 | - | - | - | 0.11 |
| Chloride (as $BaCl_2$), wt. % max. 1/ | 0.0075 | 0.0075 | 0.0075 | 0.0075 | - | 0.0075 |
| Grit, wt. % max. | 0.05 | 0.05 | 0.05 | 0.05 | - | 0.05 |
| Heavy metals, wt. % max. | 0.05 | - | - | - | - | - |
| Moisture, wt. % max. | 0.20 | 0.10 | 0.20 | 0.20 | 0.05 | 0.10 |
| Insoluble matter, wt. % max. | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 |



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TABLE II. Physical and chemical requirements for barium nitrate – Continued.

| Property | Requirements | | | | | |
|---|---|---|---|---|---|---|
| | Class 1 | Class 2 | Class 3 | Class 4 | Class 5 | Class 6 |
| Hygroscopicity, wt. % max. 2/ | - | - | - | - | 0.05 | - |
| pH | 5.0 - 8.0 | 5.0 - 8.0 | 5.0 - 8.0 | - | 5.0 - 8.0 | 5.0 - 8.0 |

Note:
1/ When alternate test method is used to verify compliance the sample shall be not more opaque than a solution prepared to the above concentration.
2/ This requirement only applicable when specified by the purchasing agency (see 6.2.h).

TABLE III. Granulation.

| U.S. Alternative Sieve Number | Granulation requirement (% thru) 1/ | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Class 1 | | Class 2 | Class 3 | Class 4 | Class 5 | Class 6 | |
| | Gran. A | Gran. B | | | | | Gran. A | Gran. B |
| 30 | - | - | 99.5 min. | - | - | - | - | - |
| 50 | - | - | - | 98.0 min. | - | - | 99.0 min. | - |
| 60 | - | - | - | - | - | 99.0 min. | - | 99.0 min. |
| 100 | 99.9 min. | 99.9 min. | - | 80.0 min. | - | 75.0 max. | 40.0 min. | - |
| 120 | - | - | - | - | 98.0 min. | - | - | - |
| 140 | 40.0 min. | 85.0 min. | - | - | - | - | - | - |
| 200 | 25.0 max. | 35.0± 10.0 | 2.0 max. | - | - | 5.0 max. | 75.0 max. | - |
| 230 | - | - | - | - | 50.0± 10.0 | - | - | - |
| 325 | - | 20.0 max. | - | - | 30.0 max. | - | - | - |

Note:
1/ U.S. Alternative Sieves shall be in accordance with ASTM E11.

3.3.1 Average particle diameter (Class 2). In addition to conforming to the granulation requirements specified in Table III, the average particle diameter of Class 2 material shall be not less than 100 microns and not greater than 180 microns.



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3.3.2 Average particle diameter (Class 6, Granulation B). In addition to conforming to the granulation requirements specified in Table III, the average particle diameter of Class 6 Granulation B material shall be not greater than 20 microns.

3.4 Workmanship. Barium nitrate shall be uniform and free from visible impurities and foreign material such as dirt, rust, paint, wooden particles or any other foreign objects.

4. VERIFICATION

TABLE IV. Requirements/verification cross reference matrix.

| Method of verification<br>1 – Analysis<br>2 – Demonstration | | 3 – Examination<br>4 – Test | | | | Classes of verification<br>A – First article<br>B – Conformance | | |
|---|---|---|---|---|---|---|---|---|
| Section 3 Requirement | Description | Verification Methods | | | | Verification Class | | Section 4 Verification |
| | | 1 | 2 | 3 | 4 | A | B | |
| 3.1.1 | First article | X | | X | | X | | 4.2 |
| 3.1.2 | Conformance | X | | X | | | X | 4.3 |
| 3.2 | Color | | | X | | X | X | 4.4.1 |
| 3.3 | Barium nitrate | X | | | | X | X | 4.4.2 |
| 3.3 | Strontium | X | | | | X | X | 4.4.3 |
| 3.3 | Calcium (Ca) | X | | | | X | X | 4.4.3 |
| 3.3 | Iron and aluminum | X | | | | X | X | 4.4.3 |
| 3.3 | Calcium and magnesium (oxides) | X | | | | X | X | 4.4.3 |
| 3.3 | Sodium | X | | | | X | X | 4.4.3 |
| 3.3 | Chloride | X | | | | X | X | 4.4.4 |
| 3.3 | Grit | X | | | | X | X | 4.4.5 |
| 3.3 | Heavy metals | X | | | | X | X | 4.4.6 |
| 3.3 | Moisture | X | | | | X | X | 4.4.7 |
| 3.3 | Insoluble matter | X | | | | X | X | 4.4.8 |
| 3.3 | Hygroscopicity | X | | | | X | X | 4.4.9 |
| 3.3 | pH | X | | | | X | X | 4.4.10 |
| 3.3 | Granulation | X | | | | X | X | 4.4.11 |
| 3.3.1 & 3.3.2 | Average particle diameter | | | X | | X | X | 4.4.12 |
| 3.4 | Workmanship | | | X | | X | X | 4.4.13 |

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see paragraph 4.2).
b. Conformance inspection (see paragraph 4.3).



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4.2 First article inspection. When specified (see 6.2), a sample shall be subjected to first article verification in accordance with 4.3.3. Samples collected in accordance with 4.3.2 shall be submitted for examination. Samples shall be produced using the same process procedures, material supply sources, production equipment, and production line as will be used in regular production.

4.2.1 Rejection. If any sample fails to comply with any of the applicable requirements the first article quantity shall be rejected.

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Inspection lots and batches shall comply with the MIL-STD-1916, paragraph "Formation and identification of lots or batches." Each batch shall consist of that quantity of barium nitrate that has been subjected to the same chemical and physical mixing process intended to make the final product homogeneous. A lot shall consist of one or more acceptable batches of barium nitrate produced by one manufacturer in accordance with one specification and revision under one continuous set of operating conditions.

4.3.2 Barium nitrate sampling. Sampling for inspection shall be performed at either the supplier location where the barium nitrate is packaged, or at the manufacturer's receiving facility where the material is unpacked for use in the next higher sub-assembly or formulation.

4.3.2.1 Sampling at the supplier. Samples shall be collected from the beginning, middle, and end of every batch in every lot in accordance with ASTM E300. A composite sample of not less than 500 grams shall be formed which is assumed to be representative of that batch. All composite samples shall be verified in accordance with 4.4. If any sample fails to comply with any applicable requirements the batch shall be rejected.

4.3.2.2 Sampling at the receiving facility. Barium nitrate packaged in shipping containers (e.g., drums, boxes, etc.) shall be sampled using Verification Level II of MIL-STD-1916. Each shipping container shall be sampled in three locations in accordance with ASTM E300 using a thief stick. This material shall be placed in a secondary airtight container and mixed to form a composite sample of not less than 500 g for each barium nitrate lot received. There shall be no reduced sampling. Every lot received shall be tested in accordance with 4.3.3. If any sample fails to meet any test requirement, the lot represented by the sample shall be rejected.



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| 4.3.3 Barium nitrate – First Article and Classification of Characteristics. | | | | |
|---|---|---|---|---|
| Category | Examination or Test | Verification Level | Requirement Paragraph | Inspection Method |
| Critical | None defined | | | |
| Major | | | | |
| 101 | Color | VL-II and 4.3.2 | 3.2 | 4.4.1 |
| 102 | Barium nitrate | 4.3.2 | 3.3 | 4.4.2 |
| 103 | Strontium | 4.3.2 | 3.3 | 4.4.3 |
| 104 | Calcium (Ca) | 4.3.2 | 3.3 | 4.4.3 |
| 105 | Iron and aluminum | 4.3.2 | 3.3 | 4.4.3 |
| 106 | Calcium and magnesium (oxides) | 4.3.2 | 3.3 | 4.4.3 |
| 107 | Sodium | 4.3.2 | 3.3 | 4.4.3 |
| 108 | Chloride | 4.3.2 | 3.3 | 4.4.4 |
| 109 | Grit | 4.3.2 | 3.3 | 4.4.5 |
| 110 | Heavy metals | 4.3.2 | 3.3 | 4.4.6 |
| 111 | Moisture | 4.3.2 | 3.3 | 4.4.7 |
| 112 | Insoluble matter | 4.3.2 | 3.3 | 4.4.8 |
| 113 | Hygroscopicity | 4.3.2 | 3.3 | 4.4.9 |
| 114 | pH | 4.3.2 | 3.3 | 4.4.10 |
| 115 | Granulation | 4.3.2 | 3.3 | 4.4.11 |
| 116 | Average particle diameter | 4.3.2 | 3.3.1 or 3.3.2 | 4.4.12 |
| Minor | | | | |
| 201 | Workmanship | VL-II and 4.3.2 | 3.4 | 4.4.13 |

4.4 Test methods and procedures. Unless otherwise specified, all chemicals and reagents shall be reagent grade or better, and water shall be distilled, deionized, or equivalent, conforming to ASTM D1193 Type I or Type II.

4.4.1 Color. Verify by visual examination that color requirement is met.

4.4.2 Barium nitrate. Determine barium nitrate content in accordance with either of the following procedures.

4.4.2.1 Barium nitrate purity.

4.4.2.1.1 Reagents.

a. Sulfuric acid, concentrated, 95-98 wt. %.
b. Hydrochloric acid, concentrated, 34-37 wt. %.
c. Sulfuric acid, wt. %, made by diluting 3.0 ± 0.1 mL of 95-98 wt. % sulfuric acid to 100 mL with water. This shall be dispensed from a burette or dropping funnel.



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4.4.2.1.2 Procedure. Perform the analysis in duplicate, i.e. on two separate, 0.5 g aliquots of barium nitrate.

a. Transfer a 0.5000 g ± 0.0100 g sample to a 600 mL beaker. Add 300 mL of water and 2-3 mL of hydrochloric acid.

b. Heat over a hotplate to dissolve the sample.

c. Remove sample from heat, and add dropwise with rapid stirring 5.0 ± 0.5 mL of 5% sulfuric acid at a rate of about 2 drops per second.

d. Cover the beaker with a watch glass and return it to the hotplate.

e. Digest for at least two hours. Note that the liquid may become superheated, but still not visibly boil unless disturbed. Towards the end of digestion, rinse off the watch glass into the beaker, and make up the beaker contents to 300 mL with water. Turn off the hotplate and allow the beaker to cool until it is still warm but cool enough to handle.

f. Weigh to the nearest 0.0001 g a 30-mL Gooch crucible equipped with an appropriately sized glass fiber filter with 2.5 micron porosity or finer. The crucible (but not the filter) shall have been previously rinsed with water, ignited for 1-2 hours at 600ºC and cooled prior to weighing.

g. Insert the crucible in a vacuum filtration apparatus, apply vacuum, and make sure the pad is properly seated. Decant the sample solution through the filter. Transfer the precipitate into the crucible with a stream of water from a squeeze bottle.

h. Wipe down and loosen any precipitate off the beaker walls with a rolled up 4x4-cm piece of ashless filter paper held with tweezers. Wash down the beaker contents into the crucible with water. Repeat until all the precipitate is transferred to the crucible. Elute the crucible with two 30 mL portions of water and place the rolled up piece of filter paper into the crucible.

i. Ignite the crucible and contents in a muffle furnace at 600ºC for 5 hours. Remove crucible from furnace, cool on a heat-proof surface and then to room temperature in a desiccator.

j. Calculate the percent barium nitrate for each of the duplicates as follows:

$$\text{Percent barium nitrate} = \frac{111.978 \times (A - B)}{C}$$

Where:

A = gross weight of crucible, filter, and precipitate, grams
B = tare weight of crucible and filter, grams
C = sample weight, grams

k. Calculate the mean of the two duplicates, round to the nearest 0.1 wt. % and report as percent barium nitrate.

4.4.2.2 Barium nitrate purity (alternate). Approximately 10 g portion of the sample shall be weighed to the nearest 0.1 mg in a 1-L beaker or volumetric flask, dissolve in about 800 mL of water and dilute to 1 liter. Pipet a 50-mL aliquot into a 400-mL beaker and dilute to about 225 mL with water. Add 6 mL of hydrochloric acid and heat to about 80˚C. While stirring, add 10 mL of ammonium acetate solution (40%), 25 mL of potassium dichromate solution (10%), and 10 grams of reagent-grade urea. Cover with a watch glass, heat to boiling and boil moderately until precipitate settles on the bottom of the beaker (this will take about 20 to 30 minutes), then continue to boil moderately for 90 minutes. Midway during the 90-minute boiling



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period, wash down the cover lid with water and add sufficient hot water to bring up the volume to about 250 mL. At the end of the heating period, filter the solution (while hot) through a tared sintered glass crucible of fine porosity. Transfer the precipitate to the crucible with potassium dichromate wash solution (made by diluting 40-mL of 10% potassium dichromate solution to 1 liter with water) and finally wash the precipitate four times with water. Dry the crucible at 120˚C for 1 hour, cool in a desiccator and weigh. Calculate the percent barium nitrate as follows:

$$\text{Percent barium nitrate} = \frac{103.16 \times A}{W}$$

Where: A = weight of precipitate, grams
W = weight of sample, in aliquot, grams

4.4.3 Strontium, calcium, iron, aluminum, magnesium, and sodium by Inductively Coupled Plasma – Optical Emission Spectroscopy (ICP-OES). Class A glassware (see 6.7) shall be used throughout this method.

4.4.3.1 Reagents. It is advisable to use commercially available stock solutions with certified concentrations traceable to primary standards (National Institute of Standards and Technology or international measurement standards). Stock solutions shall be replaced after one year or in accordance with manufacturer recommendations. Alternately, stock solutions may be prepared in-house from high purity metals or their salts; these materials shall be Reagent grade or better. The maximum recommended shelf life is one year from date of initial preparation when kept tightly sealed at 5-25˚C. Common commercially available stock concentrations are as follows:

a. Certified Na stock solution, 100 ± 2 mg/L in 0-3% (v/v) $HNO_3$
b. Certified Mg stock solution, 100 ± 2 mg/L in 0-3% (v/v) $HNO_3$
c. Certified Al stock solution, 100 ± 2 mg/L in 0-3% (v/v) $HNO_3$
d. Certified Ca stock solution, 100 ± 2 mg/L in 0-3% (v/v) $HNO_3$
e. Certified Fe stock solution, 100 ± 2 mg/L in 0-3% (v/v) $HNO_3$
f. Certified Sr stock solution, 100 ± 2 mg/L in 0-3% (v/v) $HNO_3$

Additional necessary reagents are:

g. De-ionized water, conductivity <0.1 µS/cm; LC-MS (liquid chromatography – mass spectrometry) grade water may be used instead.
h. Nitric Acid ($HNO_3$), concentrated, 67-70%, trace-metal grade
i. Nitric Acid ($HNO_3$), 3%, made by diluting 22 ± 2mL of concentrated nitric acid to 500 mL with deionized water; 500 mL is the minimum recommended volume of 3% nitric acid required for analysis.
j. Compressed argon, as per ICP-OES manufacturer recommendations

4.4.3.2 Preparation of reference (standard) solutions. Prepare 1 mg/L, 5 mg/L, and 10 mg/L standards. If using the stock concentrations listed in 4.4.3.1 then use the following steps.



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a. Pipette 1 mL of each certified stock solution into a 100-mL volumetric flask. Fill to the mark with 3% nitric acid and stopper the flask. Invert several times to mix. This is the 1 mg/L standard.

b. Pipette 5 mL of each certified stock solution into a 100-mL volumetric flask. Fill to the mark with 3% nitric acid and stopper the flask. Invert several times to mix. This is the 5 mg/L standard.

c. Pipette 10 mL of each certified stock solution into a 100-mL volumetric flask. Fill to the mark with 3% nitric acid and stopper the flask. Invert several times to mix. This is the 10 mg/L standard.

d. Prepare a fourth flask of 3% nitric acid and retain as the blank solution.

4.4.3.3 Sample preparation.

a. Weigh 1000 ± 50 mg of sample to the nearest 1 mg and transfer to a 100-mL volumetric flask.

b. Add approximately 90 mL of the 3% nitric to the flask and place the stopper on the flask.

c. Allow the sample to completely dissolve. Agitation by means of rotary shaker or ultrasonic bath may be used if necessary.

d. Fill to the mark with 3% nitric and invert several times to mix.

4.4.3.4 Instrument preparation. Prepare the instrument according to manufacturer instructions. It is recommended to perform all measurements in radial mode. Instrument response for each analyte shall be such that the response of the 1 mg/L standard is >10 times that of the blank, and the response of the 10 mg/L solution does not saturate the detector. Ensure a steady baseline with no drift; measure the response of the 5 mg/L standard at 5-minute intervals until successive measurements differ by no more than 5 %. Allow for adequate flush time of the system between standard and sample solutions to ensure there is no carryover. Calibrate the system for all six analytes using the standard solutions and blank. Analyze the samples in the same run, recording the results in mg/L. The wavelengths used for each element are as follows:

TABLE V. Wavelengths for ICP-OES element analysis.

| Element | Wavelength(nm) |
|---|---|
| Sr | 407.771 |
| Ca | 393.366 |
| Fe | 259.940 |
| Al | 167.079 |
| Mg | 279.553 |
| Na | 589.592 |

4.4.3.5 Analysis.

a. For each element, use the prepared standards and the blank to generate a four-point calibration line, which relates concentrations in mg/L to instrument response. Force the line through zero concentration. The line shall have a best fit value of ($R^2$) of not less than 0.95.



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b. Aspirate each sample for the time required by the instrument to achieve a stable response. Measure the response and calculate the concentration of each element from its corresponding calibration line, recording the results in mg/L.

c. If the calculated concentration is higher than 10 mg/L, perform a 10-fold dilution of the sample (e.g. 5 mL of original sample solution diluted to 50 mL with 3% nitric acid) and repeat previous step using the 10-fold-diluted sample. Multiply the result by 10 to determine the original concentration in mg/L.

d. Calculate the percent compound for direct comparison to the chemical requirements in Table II as follows. Multiply the measured concentration in mg/L for each element by the corresponding conversion factor in Table VI. Furthermore, calculate the total weight percent of aluminum oxide plus iron oxide for Classes 2, 4, and 6, and the total weight percent of calcium oxide plus magnesium oxide for Class 2.

TABLE VI. Conversion factors.

| Element | Chemical requirement | Conversion factor |
|---|---|---|
| Sr | Strontium (as Sr) | 0.0100 |
| Ca | Calcium (as Ca) | 0.0100 |
| Fe | Iron (as $Fe_2O_3$) | 0.0143 |
| Al | Aluminum (as $Al_2O_3$) | 0.0189 |
| Mg | Magnesium (as MgO) | 0.0166 |
| Ca | Calcium (as CaO) | 0.0140 |
| Na | Sodium (as Na) | 0.0100 |

4.4.4 Chloride (for all Classes except 5). Determine chloride content in accordance with either of the following methods.

4.4.4.1 Chloride by precipitation. Dissolve a 100 gram sample in 800 mL of hot water, filter through a Whatman No. 40 (or equivalent) filter paper and quantitatively transfer any insoluble matter to the paper with hot water. Save the insoluble material on the filter paper for the determination of grit (see 4.4.5). Add 5 mL of nitric acid to the filtrate, heat to boiling, and add 5 mL of 5% silver nitrate solution. Continue boiling until the solution is clear, then allow the precipitate to settle in the dark at room temperature for 3 or more hours. Filter through a tared sintered glass crucible and wash with 1% nitric acid solution. Dry at 130-150˚C for one hour, cool in a desiccator, and weigh. Calculate the percent chloride (as barium chloride) as follows:

$$\text{Percent chloride (as barium chloride)} = \frac{72.7 \times A}{W}$$

Where: A = weight of precipitate, grams
W = weight of sample, grams

4.4.4.2 Chloride by turbidity (alternate). Determine chloride content in accordance with ACS Reagent Chemicals, Chloride Method for barium nitrate.



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4.4.5 Grit (for all Classes except 5). Wash the insoluble matter remaining in the filter paper from the chloride determination into a beaker with a minimum amount of hot water. Add 10 mL of nitric acid and 30 mL of hydrochloric acid. Heat the mixture on a steam bath for 1 hour, cool, and decant the supernatant liquid. Wash any insoluble material with water, by decantation, transfer to a filter paper, and continue washing until free from acid. Transfer the insoluble reside to a US Alternative No. 120 sieve complying with ASTM E11, and by means of a jet of water, wash until no more material will pass through the sieve. Rinse the sieve and residue with alcohol and then with ether, and allow to dry at room temperature. Transfer any residue on the sieve to a tared weighing bottle, dry at 100-105˚C, for one hour, cool in a desiccator, and weigh. Calculate the weight of the residue to percent grit. Transfer the material to a glass slide and rub between two glass slides. Determine the presence of grit by scratching noise and scratches on glass slides.

4.4.6 Heavy metals (for Class 1). Dissolve 25 grams of the sample in distilled water. Add a few drops of hydrochloric acid and then pass in hydrogen sulfide for two or three minutes. Inspect for the formation for a precipitate. Render the solution ammoniacal and allow to stand for three hours. Again inspect for the formation of precipitate. The appearance of a precipitate in the acid solution or alkaline solution is indicative of the presence of heavy metals.

4.4.7 Moisture. Transfer a ten gram sample to a weighing bottle, and weigh the weighing bottle plus sample to the nearest 0.1 g. Heat at 100-105˚C for 3 hours, cool in a desiccator, and weigh. Calculate the percent moisture as follows:

$$\text{Percent moisture} = \frac{100 \times A}{W}$$

Where: A = loss in weight, grams
W = weight of sample, grams

4.4.8 Insoluble matter. Transfer a 10 gram sample to a 400-mL beaker, add 200 mL of hot water and stir to dissolve. Filter through a tared sintered glass crucible of medium porosity, transfer and wash thoroughly with hot water. Dry the crucible at 100-105˚C for one hour, cool in a desiccator and weigh. Calculate the percent insoluble matter as follows:

$$\text{Percent insoluble matter} = \frac{100 \times A}{W}$$

Where: A = weight of residue, grams
W = weight of sample, grams

4.4.9 Hygroscopicity (for Class 5). The method for determination of the hygroscopicity shall be as directed by the contracting agency (see 6.2).

4.4.10 pH. Dissolve a one gram sample in 20 mL of water that has been boiled, cooled, and verified neutral using a calibrated pH meter. Determine the pH with a pH meter which has been calibrated with standard buffer solutions with nominal pH of 4, 7, and 10.



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4.4.11 Granulation. Nest the specified sieves (see 6.3) complying with ASTM E11 on a bottom pan. Place a weighed portion of 100 grams of the sample on the upper sieve, cover, and shake for five minutes by means of a mechanical shaker geared to produce 268 – 315 gyrations and 150 ± 10 taps of the striker per minute. Weigh the amounts retained by the sieves, and calculate to percent as required.

4.4.12 Average particle diameter (for Class 2 and Class 6 Granulation B). The average particle diameter shall be determined in accordance with ASTM B330. The density of barium nitrate is 3.244 g/cm$^3$.

4.4.13 Workmanship. Take approximately 100 to 200 grams of the material and spread the composition out on a clean white surface. The sample shall be examined by personnel with normal vision or normal corrected vision. The sample shall be uniform and free from lumps, visible impurities, foreign matter or other defects. Failure to meet the workmanship inspection or during following production processes shall result in batch or lot rejection.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Barium nitrate is intended for the following uses:

| | | |
|---|---|---|
| Class 1 | - | for use in the manufacture of priming compositions |
| Class 2 | - | for use in the manufacture of photoflash compositions |
| Class 3 | - | for use in the manufacture of propellants |
| Class 4 | - | for use in the manufacture of special compositions |
| Class 5 | - | for use in the manufacture of incendiary mixtures |
| Class 6 | - | for use in the manufacture of pyrotechnic compositions |

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Specific issue of individual documents referenced (see 2.1 and 2.2).



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c. Provisions for submission/resubmission of first article samples.
d. Quantity required and delivery schedules.
e. Class and Granulation required (see 1.2 and 3.3).
f. Levels of preservation and packing (see 5.1).
g. Item hazard classification.
h. Hygroscopicity if required (see Table II Note 1), and test method (see 4.4.9).
i. Propellant and Explosive Data sheets (see 6.3).

6.3 Consideration of data requirements. When this specification is used in a procurement contract which incorporates DD Form 1423, contract data requirements list, acceptance and description sheets (for explosives) are required for each lot. These sheets should be prepared and submitted by the contractor in accordance with MIL-STD-1171 or otherwise as directed by the contract.

6.4 Technical agency. US Army Combat Capability Development Command Armaments Center is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: DEVCOM-AC, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000.

6.5 Approval of equivalent test methods. Prior approval of the Government Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted for consideration through the Contracting Officer to the Technical Agency (see 6.4). This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.6 Submission of alternative conformance inspection provisions. Unless otherwise specified in the contract, proposed alternative conformance inspection provisions should be submitted by the contractor for evaluation by the Technical Agency responsible for preparation of this specification (see 6.4).

6.7 Class A glassware. Class A glassware should meet the requirements of ASTM E694 “Standard Specification for Laboratory Glass Volumetric Apparatus” and ASTM E542 “Standard Practice for Calibration of Laboratory Volumetric Apparatus.”

6.8 Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.



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6.9 Changes incorporated. This specification was updated from MIL-B-162D to MIL-DTL-162E with Amendment 2 as part of the Munitions Chemical Specification Modernization (MCSM) program. All test procedures were verified and updated as needed to produce valid results. Alternate test methods have been added. Reference documents have been verified, updated, and hyperlinked to sources. Packaging instructions have been changed. Sampling procedures have been updated. Details of these changes are documented in the ECPs and ERR listed below.

| Specification ID | ECP / ERR |
|---|---|
| MIL-B-162D | - |
| MIL-DTL-162E | Y21Q2007 |
| MIL-DTL-162E w/ Amendment 1 | Y23Q2005 |
| MIL-DTL-162E w/ Amendment 2 | Y24Q3001 |

6.10 Subject term (key word) listing.

Energetic material
Tracer ingredient
Primer ingredient
Munitions Chemical Specification Modernization (MCSM)

6.11 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodians:
Army-AR
Navy-AS

Preparing activity
Army-AR
(Project 6810-2025-003)

Review activities:
Army-EA, MD, MI
Navy-OS
DLA-GS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



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